Project No.:	01-26-020 (26766)	**RFI No.:**	RFI-126020-012
Date:	August 25, 2026	**Discipline:**	Piping
Project Name:	26766 – BA2107 TLE End-of-Life Replacement		

Subject: ASC Fig. 40 Riser Clamps – Level Bridle Piping

Specifications/Documents/Drawings Referenced or Impacted:

Isometric FA2107-TRIM Rev. 1.1, Sheets 1 and 2, Note 9

ASC Fig. 40 Riser Clamp – Standard, data sheet

ASC Fig. 261 Extension Pipe or Riser Clamp, data sheet

Question or Description of Issue:

Note 9 of isometric FA2107-TRIM Rev. 1.1, Sheets 1 and 2, specifies ASC Fig. 40 riser clamps on the FA-2107 level bridle piping. Note 9 states:

> *"REMOVE INSULATION AND INSTALL CS ASC RISER CLAMP FIG.40. RISER SHOULD REST ON STEEL. RE-INSULATE PIPING TO MATCH EXISTING."*

The 4" Fig. 40 clamps are quoted at $822.39 each and are not available until early November. ASC originally quoted Fig. 261 clamps, which are available from stock but carry a lower load rating than the specified Fig. 40. Published ratings for both figures are given in the attached data sheets.

The ASC data sheet for the Fig. 40 also states that load is carried by shear lugs welded to the pipe. Shear lugs are not identified or detailed on the scope drawings, and Note 9 instead indicates that the riser is to rest on steel.

Welding shear lugs to the FA-2107 trim piping (4"/2", pipe class 15CS03C, 1950 psig at 650°F, ASME B31.3 normal fluid service) would introduce additional welding, examination and hydrotest requirements beyond those currently shown.

Originator's Recommended Solution:

Please review the required support and restraint loads for the level bridle piping and advise whether ASC Fig. 261 clamps are acceptable in lieu of the specified Fig. 40. If the Fig. 261 is not acceptable, please advise whether a field-fabricated support such as angle iron with a tight U-bolt or similar arrangement may be used in order to avoid welding to the existing piping. Please also confirm whether shear lugs are required and, if so, issue the applicable detail together with the associated examination and test requirements.

Urgency: Medium (3 days)

Originator:

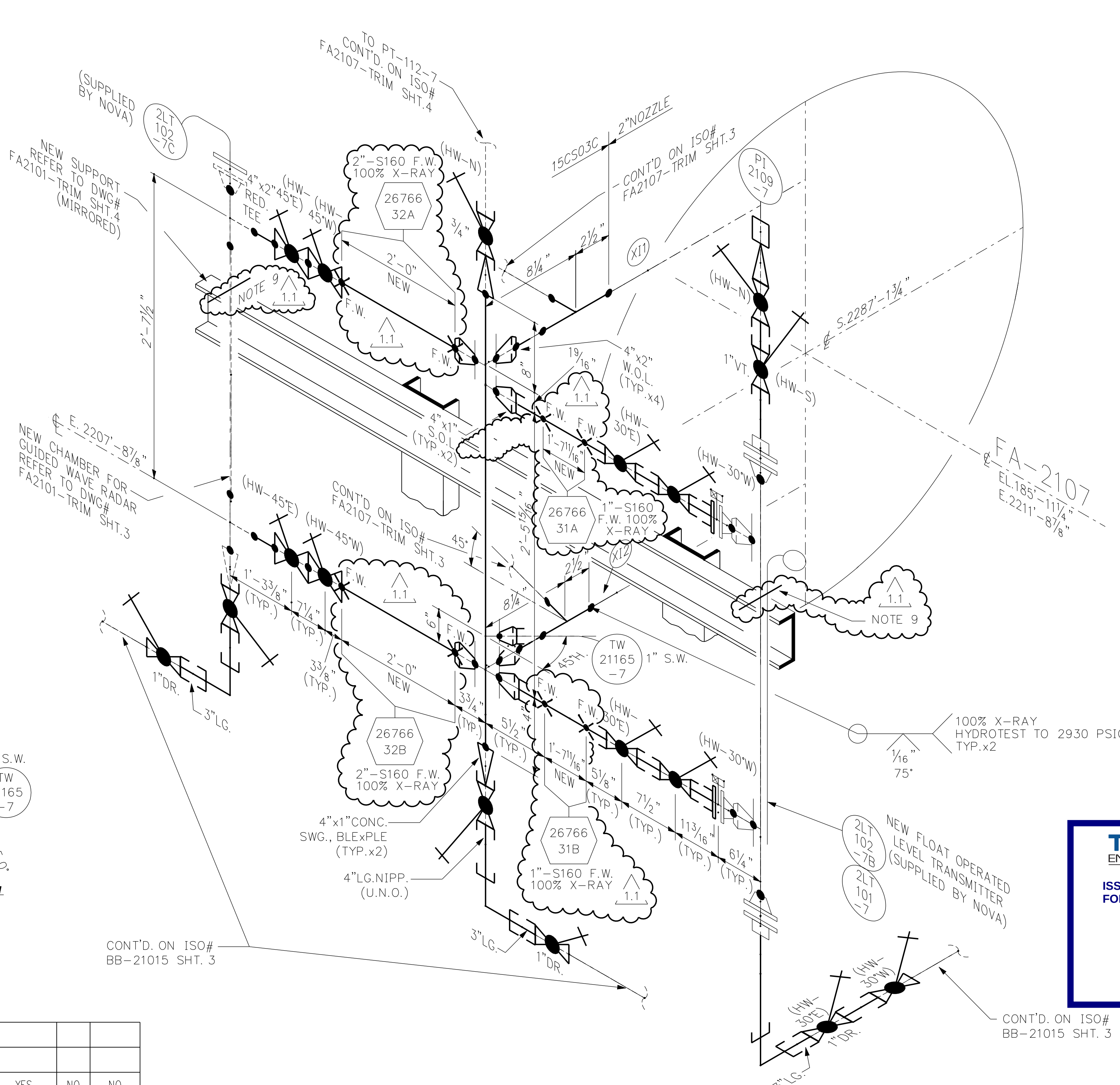
Nitin Joshi		PC	August 25, 2026
Name	Signature	Title	Date

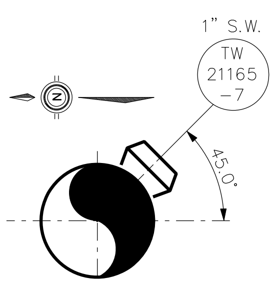
Response:

Respondent:

Name	Signature	Title	Date

NOTE: PIPING SPOOLS WERE PRE-FABRICATED.
CONTRACTOR TO INSTALL IN FIELD AS SHOWN.





DETAIL #1

PIPE

	REV.	ITEM	QUANTITY	NOMINAL DIAMETER	ENG. CODE	SAP CODE	DESCRIPTION
*		1	5'	4"	PI01CS72BW01		PIPE BE S120 SMLS ASTM A106 GR. B
*		2	3'	2"	PI01CS74BW01		PIPE S160 SMLS BE ASTM A106 GR. B
*		3	2'	1"	PI01CS74SW01		PIPE S160 SMLS PE ASTM A106 GR. B

FITTINGS

	REV.	ITEM	QUANTITY	MAIN SIZE	RED SIZE	ENG. CODE	SAP CODE	DESCRIPTION
		4	3	1"	N/A	EB04CS60SW02		ELBOW 90 6000# SW ASTM A105N
		5	2	1"	N/A	FL43CS15RF02		FLG SW S160 BORE 1500# RF ASTM A105N
		6	3	1"	N/A	NI03CS74PP01		NIPPLE 3" LG S160 PBE SMLS ASTM A106 GR B
		7	10	1"	N/A	NI04CS74PP01		NIPPLE 4" LG S160 PBE SMLS ASTM A106 GR B
		8	3	4"	1"	OL03CS60SW02		SOCKOLET 6000# ASTM A105N
		9	4	4"	2"	OL05CS74BW02		WELDOLET S160 ASTM A105N
		10	4	1"	N/A	PL04CS60TD02		PLUG 6000# THRD ROUND HEAD ASTM A105N
*		11	1	4"	1"	SC72CS72BP01		SWAGE NIP. CON. S120 BLE/S160 PSE ASTM A234 GR. WPB
*		12	1	4"	3/4"	SC72CS72BP01		SWAGE NIP. CON. S120 BLE/S160 PSE ASTM A234 GR. WPB
*		13	1	1"	1/2"			SWAGE CON S160 TBE ASTM A234 GR. WPB
*		14	1	1/2"	1/2"			CPLG FULL 6000# THRD ASTM A105N
		15	2	2"	2"	TE01CS74BW01		TEE STRAIGHT S160 BW ASTM A234 GR. WPB

VALVES

	REV.	ITEM	QUANTITY	MAIN SIZE	RED SIZE	ENG. CODE	SAP CODE	DESCRIPTION
*		16	4	2"	N/A			GLOBE, Y-PATTERN, 1500# BW, CONVAL# 12G4QZ-105 (8J)
*		17	8	1"	N/A			GLOBE, Y-PATTERN 1500# SW, CONVAL# 12G2QZ-105 (5F)
*		18	4	1"	N/A			GLOBE, Y-PATTERN 1500# SWxTHRD, CONVAL# 12G0QZ-105 (5F)
*		19	1	3/4"	N/A			GLOBE, Y-PATTERN 1500# SW, CONVAL# 12G2QZ-105 (5F)

BOLTS

	REV.	ITEM	QUANTITY	BOLT DIA.	LENGTH	ENG. CODE	SAP CODE	DESCRIPTION
*		20	8	0.875	5.0	BT01193B700		STUD BOLT A193 B7 c/w 2 HVY HEX NUTS A194 2H

GASKETS

REV.	ITEM	QUANTITY	MAIN SIZE	ENG. CODE	SAP CODE	DESCRIPTION
	21	2	1"	GKC4615RF03		1500# 304 SPIRAL WND CS IN/OUT RINGS FLEX GRAPHITE FILLED

GAUGE AND BLEED VALVES

	REV.	ITEM	QUANTITY	MAIN SIZE	ENG. CODE	SAP CODE	DESCRIPTION
*		22	1	3/4"	IG60CSEC01F	-	GAUGE VALVE, TYPE P-2, HEX MODEL # HG473S423IJ23
*		23	1	1/2"	IG60CSTD01S	-	BLEED VALVE, TYPE-P2, HEX MODEL # HB-25IS312

PIPE

1.1

REV.	ITEM	QUANTITY	NOMINAL DIAMETER	COMMODITY CODE	MID NUMBER	DESCRIPTION
	1	4'	1"	PNNPAAAAPECRAAI6	1502957	PIPE, ASME B36.10M, SMLS, PE, ASTM A106 GR. B, SCH. 160
	2	4'	2"	PNNPAAAABECRAAI6	1079677	PIPE, ASME B36.10M, SMLS, BE, ASTM A106 GR. B, SCH. 160

BILL OF MATERIAL PER NOVA PIPE SPECIFICATION 15CS03C, REVISION 9

NOTES:

1. FIELD CHECK ALL DIMENSIONS PRIOR TO PIPE FABRICATION.
2. VENT, DRAIN & PI CONNECTIONS SHALL BE GUSSETTED AS PER STANDARD DETAIL DRAWING No. ES-PPG-8117.
3. * INDICATES ITEMS IN THE BILL OF MATERIAL THAT HAVE BEEN MANUALLY MODIFIED FROM THE BITWYSE DATABASE.
4. NOVA INSPECTION SHALL VERIFY THE INTEGRITY OF ALL EXISTING, RE-USED PIPING PRIOR TO MAKING ANY TIE-INS.
5. PIPING SHALL BE EXAMINED PRIOR TO HYDROTEST AS INDICATED FOR ASME B31.3 FLUID CATEGORY.
6. VISUAL WELD INSPECTION
   - [X] NORMAL FLUID SERVICE
     5% OF FABRICATION FOR EACH WELDER SHALL BE EXAMINED AS OUTLINED IN TABLE 341.3.2.
   - [ ] CATEGORY D FLUID SERVICE
     VISUAL EXAMINATION TO EXTENT NECESSARY TO SATISFY EXAMINER.
7. FOR 2" NPS OR LESS, USE 2" THICK INSULATION & FOR 3" NPS OR GREATER, USE 2.5" THICK INSULATION.
8. SHOP HYDROTEST NEW FLOAT OPERATED LEVEL TRANSMITTER TO 3445 PSIG & ALL OTHER PIPING TO 4050 PSIG. FIELD HYDROTEST ALL PIPING TO 2930 PSIG.
9. REMOVE INSULATION AND INSTALL CS ASC RISER CLAMP FIG.40. RISER SHOULD REST ON STEEL. RE-INSULATE PIPING TO MATCH EXISTING.

TECSAR ENGINEERING INC.
ISSUED / APPROVED FOR CONSTRUCTION
06/04/2026

LICENSED PROFESSIONAL ENGINEER
R. RAHMAN
37959509
PROVINCE OF ONTARIO
Rev: 1.1
Jun-04-2026

BA-2107
HEATER #7

15CS03C	1"	TORQUE	TABLE 11	YES	NO	NO
PIPE SPEC	NPS	METHOD ES-PPG-1007	TORQUE TABLE ES-PPG-1004	INSPECTION	UT	RECORD

BOLT TIGHTENING

DES. PRESS.- 1950 PSIG.	DES. TEMP. 650 °F.	X-RAY 100 %	
HYDROTEST- NOTE 8 PSIG.	STRESS RELIEF- [ ] YES [X] NO		
TRACING	[X] STEAM	INSULATION (NOTE 7)	TYPE- IH (MW)
	[ ] ELECTRIC		THICK- IN.
PAINT [ ] YES [X] NO	PAINT CODE N/A		
Proj. No. 23346			

SIZE	LINE NUMBER	PIPE CLASS
4"/2"	FA-2107 TRIM	15CS03C
1"	FA-2107 TRIM	15CS03C

REFERENCE DRAWINGS:
P&ID- D-20006-U
PLAN- MODEL

REV.	DATE	DESCRIPTION	DRN.	CHK.	APP.
1.1	MAY 2026	ISSUED FOR CONSTRUCTION – PROJ. # 26766	MG	DG	
1	31/01 2013	APPROVED FOR CONSTRUCTION – CWP-G07 (S/D)	SB		
A	07/01 2013	ISSUED FOR APPROVAL – CWP-G07 (S/D)	SB		

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DWG. NO. FA2107-TRIM SHT. 1 OF REV. 1.1

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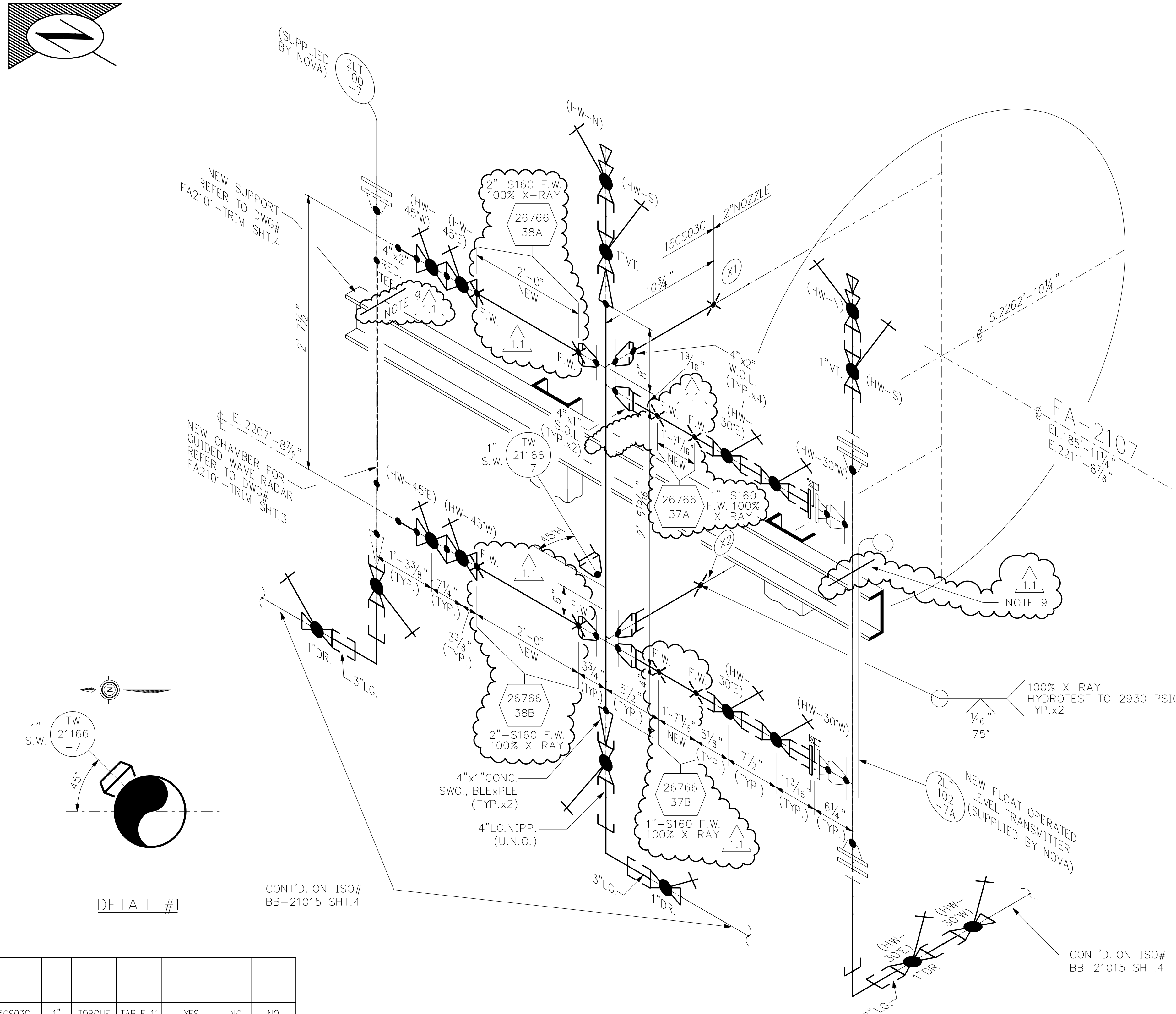

PIPE

	REV.	ITEM	QUANTITY	NOMINAL DIAMETER	ENG. CODE	SAP CODE	DESCRIPTION
*		1	5'	4"	PIO1CS72BWO1		PIPE BE S120 SMLS ASTM A106 GR. B
*		2	4'	2"	PIO1CS74BWO1		PIPE S160 SMLS BE ASTM A106 GR. B
*		3	2'	1"	PIO1CS74SWO1		PIPE S160 SMLS PE ASTM A106 GR. B

FITTINGS

	REV.	ITEM	QUANTITY	MAIN SIZE	RED SIZE	ENG. CODE	SAP CODE	DESCRIPTION
		4	3	1"	N/A	EB04CS60SW02		ELBOW 90 6000# SW ASTM A105N
		5	2	1"	N/A	FL43CS15RF02		FLG SW S160 BORE 1500# RF ASTM A105N
		6	3	1"	N/A	NI03CS74PP01		NIPPLE 3" LG S160 PBE SMLS ASTM A106 GR B
		7	11	1"	N/A	NI04CS74PP01		NIPPLE 4" LG S160 PBE SMLS ASTM A106 GR B
		8	3	4"	1"	OL03CS60SW02		SOCKOLET 6000# ASTM A105N
		9	4	4"	2"	OL05CS74BW02		WELDOLET S160 ASTM A105N
		10	5	1"	N/A	PL04CS60TD02		PLUG 6000# THRD ROUND HEAD ASTM A105N
*		11	2	4"	1"	SC72CS72BP01		SWAGE NIP. CON. S120 BLE/S160 PSE ASTM A234 GR. WPB

VALVES

	REV.	ITEM	QUANTITY	MAIN SIZE	RED SIZE	ENG. CODE	SAP CODE	DESCRIPTION
*		12	4	2"	N/A			GLOBE, Y-PATTERN, 1500# BW, CONVAL# 12G4QZ-105 (8J)
*		13	9	1"	N/A			GLOBE, Y-PATTERN 1500# SW, CONVAL# 12G2QZ-105 (5F)
*		14	5	1"	N/A			GLOBE, Y-PATTERN 1500# SWxTHRD, CONVAL# 12G0QZ-105 (5F)

BOLTS

	REV.	ITEM	QUANTITY	BOLT DIA.	LENGTH	ENG. CODE	SAP CODE	DESCRIPTION
*		15	8	0.875	5.0	BT01193B700		STUD BOLT A193 B7 c/w 2 HVY HEX NUTS A194 2H

GASKETS

REV.	ITEM	QUANTITY	MAIN SIZE	ENG. CODE	SAP CODE	DESCRIPTION
	16	2	1"	GKC4615RF03		1500# 304 SPIRAL WND CS IN/OUT RINGS FLEX GRAPHITE FILLED

PIPE

1.1

REV.	ITEM	QUANTITY	NOMINAL DIAMETER	COMMODITY CODE	MID NUMBER	DESCRIPTION
	1	4'	1"	PNNPAAAAPECRAAI6	1502957	PIPE, ASME B36.10M, SMLS, PE, ASTM A106 GR. B, SCH. 160
	2	4'	2"	PNNPAAAABECRAAI6	1079677	PIPE, ASME B36.10M, SMLS, BE, ASTM A106 GR. B, SCH. 160

BILL OF MATERIAL PER NOVA PIPE SPECIFICATION 15CS03C, REVISION 9

1.1 NOTE 9

NOTES:

1. FIELD CHECK ALL DIMENSIONS PRIOR TO PIPE FABRICATION.
2. VENT, DRAIN & PI CONNECTIONS SHALL BE GUSSETTED AS PER STANDARD DETAIL DRAWING No. ES-PPG-8117.
3. * INDICATES ITEMS IN THE BILL OF MATERIAL THAT HAVE BEEN MANUALLY MODIFIED FROM THE BITWYSE DATABASE.
4. NOVA INSPECTION SHALL VERIFY THE INTEGRITY OF ALL EXISTING, RE-USED PIPING PRIOR TO MAKING ANY TIE-INS.
5. PIPING SHALL BE EXAMINED PRIOR TO HYDROTEST AS INDICATED FOR ASME B31.3 FLUID CATEGORY.
6. VISUAL WELD INSPECTION
   - [X] NORMAL FLUID SERVICE
     5% OF FABRICATION FOR EACH WELDER SHALL BE EXAMINED AS OUTLINED IN TABLE 341.3.2.
   - [ ] CATEGORY D FLUID SERVICE
     VISUAL EXAMINATION TO EXTENT NECESSARY TO SATISFY EXAMINER.
7. FOR 2" NPS OR LESS, USE 2" THICK INSULATION & FOR 3" NPS OR GREATER, USE 2.5" THICK INSULATION.
8. SHOP HYDROTEST NEW FLOAT OPERATED LEVEL TRANSMITTER TO 3445 PSIG & ALL OTHER PIPING TO 4050 PSIG. FIELD HYDROTEST ALL PIPING TO 2930 PSIG.
9. REMOVE INSULATION AND INSTALL CS ASC RISER CLAMP FIG.40. RISER SHOULD REST ON STEEL. RE-INSULATE PIPING TO MATCH EXISTING.

TECSAR ENGINEERING INC.
ISSUED / APPROVED FOR CONSTRUCTION
06/04/2026

LICENSED PROFESSIONAL ENGINEER
S. RAHMAN
37959509
PROVINCE OF ONTARIO
Rev: 1.1
Jun-04-2026

BA–2107
HEATER #7

15CS03C	1"	TORQUE	TABLE 11	YES	NO	NO
PIPE SPEC	NPS	METHOD ES-PPG-1007	TORQUE TABLE ES-PPG-1004	INSPECTION	UT	RECORD

BOLT TIGHTENING

DES. PRESS.- 1950 PSIG.	DES. TEMP. 650 °F.	X-RAY 100 %	
HYDROTEST- NOTE 8 PSIG.	STRESS RELIEF- [ ] YES [X] NO		
TRACING [X] STEAM [ ] ELECTRIC	INSULATION (NOTE 7)	TYPE- IH (MW) THICK- IN.	
PAINT [ ] YES [X] NO	PAINT CODE N/A		
Proj. No. 23346			

SIZE	LINE NUMBER	PIPE CLASS
4"/2"	FA-2107 TRIM	15CS03C
1"	FA-2107 TRIM	15CS03C

REFERENCE DRAWINGS:
P&ID- D-20006-U
PLAN- MODEL

REV.	DATE	DESCRIPTION	DRN.	CHK.	APP.
1.1	MAY 2026	ISSUED FOR CONSTRUCTION – PROJ. # 26766	MG	DG	
1	31/01 2013	APPROVED FOR CONSTRUCTION – CWP-G07 (S/D)	SB		
A	07/01 2013	ISSUED FOR APPROVAL – CWP-G07 (S/D)	SB		

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PIPE

REV.	ITEM	QUANTITY	NOMINAL DIAMETER	ENG. CODE	SAP CODE	DESCRIPTION
*	1	6	11/2"	PIOICS74SWOI		PIPE SI60 SMLS PE ASTM AI06 GR. B
	2	1'	2"	PIOICS74BWOI		PIPE SI60 SMLS BE ASTM AI06 GR. B

FITTINGS

REV.	ITEM	QUANTITY	MAIN SIZE	RED SIZE	ENG. CODE	SAP CODE	DESCRIPTION
	3	2	11/2"	11/2"	CPOICS60SWOI		CPLG FULL 6000# SW ASTM AI05N
	4	2	11/2"	N/A	EBOICS60SW02		ELBOW 45 6000# SW ASTM AI05N
	5	1	2"	N/A	EB02CS74BWOI		ELBOW 45 LR SI60 BW ASTM A234 GR WPB
	6	2	2"	11/2"	SC74CS74BPOI		SWAGE NIP. CON. SI60 BLE/PSE ASTM A234 GR. WPB

VALVES

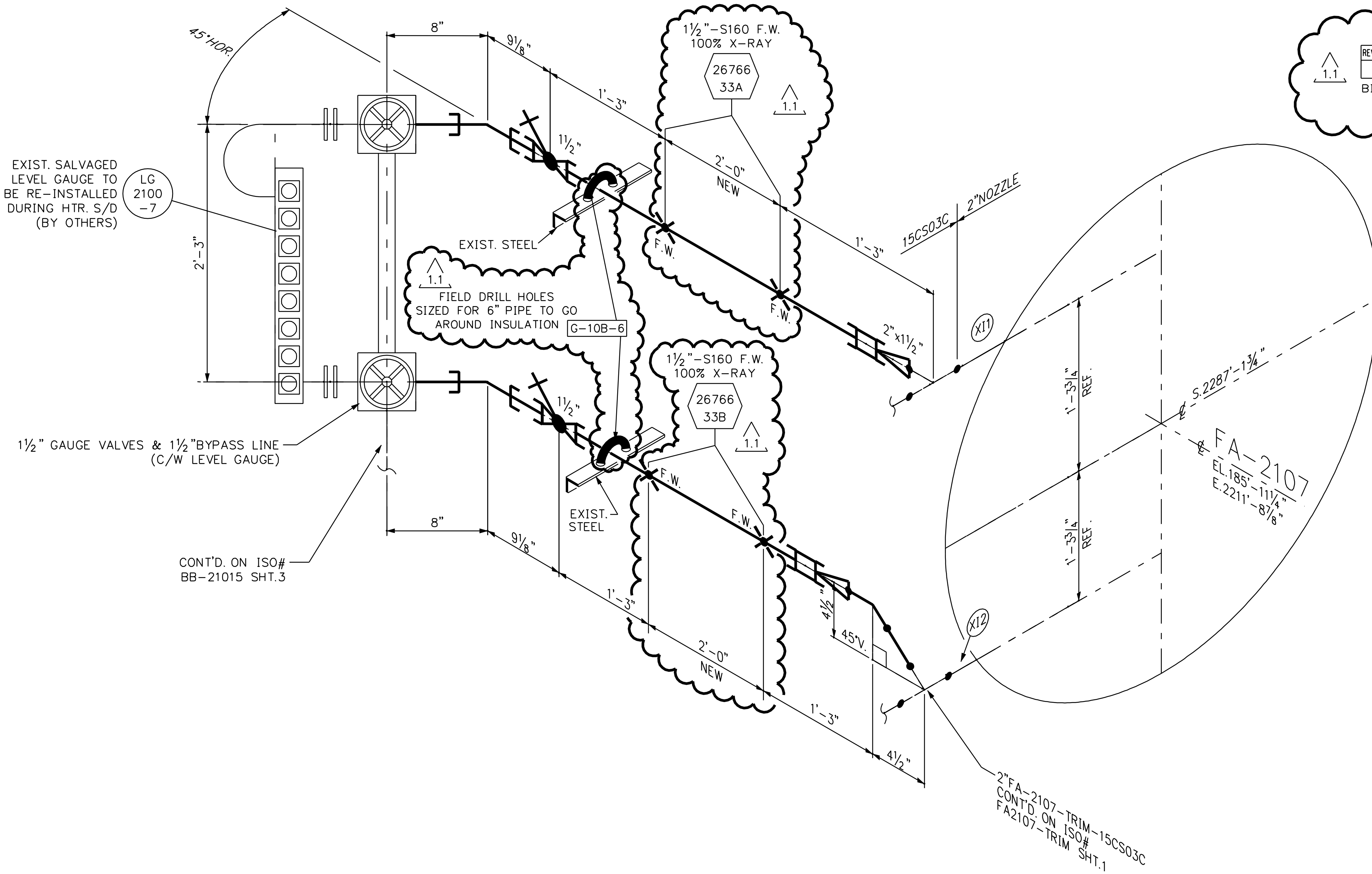
REV.	ITEM	QUANTITY	MAIN SIZE	RED SIZE	ENG. CODE	SAP CODE	DESCRIPTION
*	7	2	11/2"	N/A			GLOBE, Y-PATTERN 1500# SW, CONVAL# 12G2QZ-105 (7H)

PIPE

1.1

REV.	ITEM	QUANTITY	NOMINAL DIAMETER	COMMODITY CODE	MID NUMBER	DESCRIPTION
	1	4'	11/2"	PNNPAAAAPECRAAI6	1504466	PIPE, ASME B36.10M, SMLS, PE, ASTM AI06 GR. B, SCH. 160

BILL OF MATERIAL PER NOVA PIPE SPECIFICATION 15CS03C, REVISION 9



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FOR CONSTRUCTION
06/04/2026

NOTES:

1. FIELD CHECK ALL DIMENSIONS PRIOR TO PIPE FABRICATION.
2. VENT, DRAIN & PI CONNECTIONS SHALL BE GUSSETTED AS PER STANDARD DETAIL DRAWING No. ES-PPG-8117.
3. * INDICATES ITEMS IN THE BILL OF MATERIAL THAT HAVE BEEN MANUALLY MODIFIED FROM THE BITWYSE DATABASE.
4. NOVA INSPECTION SHALL VERIFY THE INTEGRITY OF ALL EXISTING, RE-USED PIPING PRIOR TO MAKING ANY TIE-INS.
5. PIPING SHALL BE EXAMINED PRIOR TO HYDROTEST AS INDICATED FOR ASME B31.3 FLUID CATEGORY.
6. VISUAL WELD INSPECTION
   - [X] NORMAL FLUID SERVICE
     5% OF FABRICATION FOR EACH WELDER SHALL BE EXAMINED AS OUTLINED IN TABLE 341.3.2.
   - [ ] CATEGORY D FLUID SERVICE
     VISUAL EXAMINATION TO EXTENT NECESSARY TO SATISFY EXAMINER.
7. SHOP HYDROTEST NEW FLOAT OPERATED LEVEL TRANSMITTER TO 3445 PSIG & ALL OTHER PIPING TO 4050 PSIG. FIELD HYDROTEST ALL PIPING TO 2930 PSIG.

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RAHMAN
37959509
Rev: 1.1
PROVINCE OF ONTARIO
Jun-04-2026

BA-2107
HEATER #7

DES. PRESS.- 1950 PSIG.	DES. TEMP. 650°F.	X-RAY 100 %	SIZE	LINE NUMBER	PIPE CLASS	1.1	MAY 2026	ISSUED FOR CONSTRUCTION – PROJ. # 26766	MG / DG
HYDROTEST- NOTE 8 PSIG.	STRESS RELIEF- [ ] YES [X] NO		1½"/1"	FA-2107 TRIM	15CS03C	1	31/01 2013	APPROVED FOR CONSTRUCTION – CWP-G07 (S/D)	SB / JG / CB
TRACING [X] STEAM [ ] ELECTRIC	INSULATION	TYPE- IH (MW) THICK- 2 IN.	REFERENCE DRAWINGS:			A	07/01 2013	ISSUED FOR APPROVAL – CWP-G07 (S/D)	SB / JG / CB
PAINT [ ] YES [X] NO	PAINT CODE N/A		P&ID- D-20006-U						
Proj. No. 23346			PLAN- MODEL			REV.	DATE	DESCRIPTION	DRN. / CHK. / APP.

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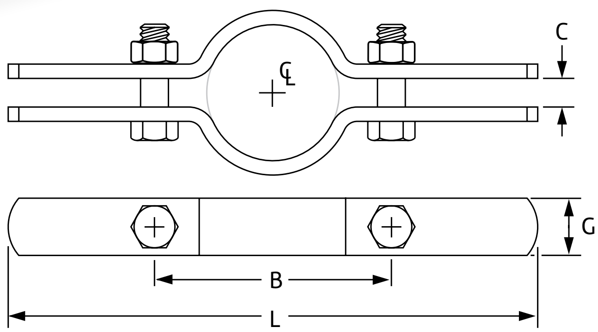
DWG. NO. FA2107-TRIM SHT. 3OF REV. 1.1

ASC Engineered Solutions

Extension Pipe or Riser Clamp
Fig. 261 (Formerly Afcon Fig. 400)

c UL us LISTED



Size Range: ¾" through 24"

Material: Carbon steel

Finish: ☐Plain, ☐Hot-Dip Galvanized with Zinc Plated Bolts & Nuts, ☐Epoxy Coated or ☐Painted.

Service: For support of stationary steel pipe risers, cast iron pipe or conduit. This product is not intended for use with hanger rods. For this application refer to Fig. 40 Riser Clamp.

Maximum Temperature: Plain 650° F, Galvanized and Epoxy 450° F

Approvals: Complies with Federal Specification A-A-1192A (Type 8) WW-H-171-E (Type 8), ANSI/MSS SP-69 and MSS SP-58 (Type 8). UL, ULC Listed (Sizes 1½" - 8").

Installation: Clamp is fitted and bolted preferably below a coupling, hub or welded lugs on steel pipe. Bolt torques should be per industry standards (see page 248). Clamp is designed for standard steel pipe O.D. and this must be considered in sizing the riser for other types of piping.

Ordering: Specify pipe size, figure number, name and finish.

Note: Refer to Technical Data Section of the Pipe Hanger Catalog for cast iron soil pipe data.

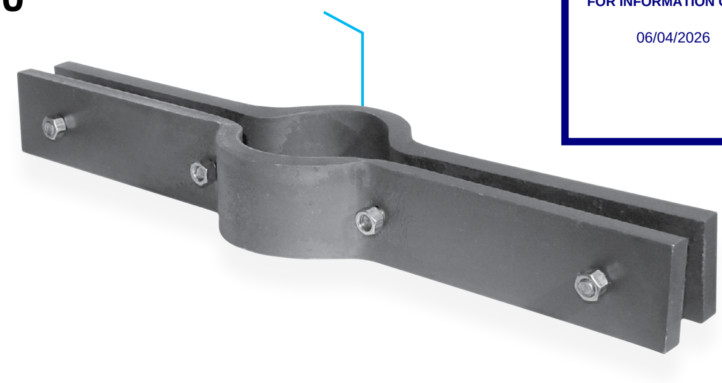
Fig. 261: Dimensions (in) • Loads (lbs) • Torque (ft-lbs) • Weight (lbs)

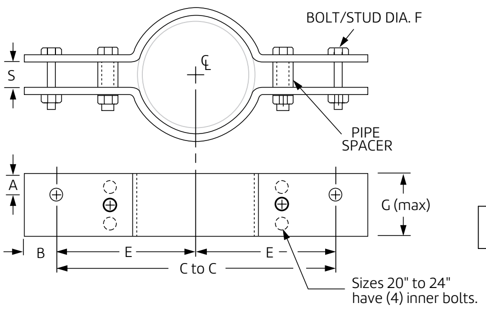
Pipe Size	Max Load	Weight	L	G Width	B	C	Bolt Diameter	Torque Values
½ & ¾	220	1.1	8⅞	1	2⅞	⅜	⅜	21
1		1.1			3⅛			
1¼	250	1.6	10		3½			
1½		1.6	10¼		3⅞			
2	300	1.7			4¼	½	7/16	32
2½	400	1.9	11¼		4¾			
3	500	1.9	11⅜		5½			
3½	600	2.3	12⅞		6½	½	½	46
4	750	2.4			7			
5	1,500	3.6	13¾	1½	8			
6	1,600	4.0	14¾		9			
8	2,500	7.6	18½		12	⅝	⅝	100
10		11.1	20¼	2	13¾			
12	2,700	16.5	22¾		15¾			
14		17.7	24		17¼			
16	2,900	30.4	26	2½	19¾	¾	¾	150
18		33.8	28		21¾			
20		35.0	30		23¾			
24	3,200	82.0	36¾	3	30	1	⅞	190

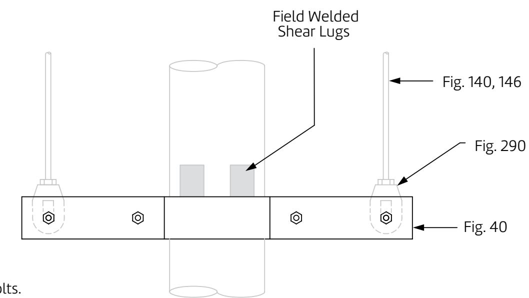
PROJECT INFORMATION	APPROVAL STAMP
Project:	☐ Approved
Address:	☐ Approved as noted
Contractor:	☐ Not approved
Engineer:	Remarks:
Submittal Date:	
Notes 1:	
Notes 2:	

ASC Engineered Solutions

Riser Clamp – Standard
Fig. 40







Size Range: 2" through 24"

Material: Carbon steel (CS), Alloy (A), or Stainless Steel (SS)

Finish: ☐ Plain or ☐ Hot-Dip Galvanized with Zinc Plated Fasteners

Maximum Temperature: Galvanized 450° F, 650° F (CS), 950° F (A) and 1,000° F (SS)

Service: Riser clamps are used for the support of vertical piping. Load is carried by shear lugs which are welded to the pipe. Shear lugs provided upon request. Local pipe wall stress evaluation available upon request.

Approvals: Complies with Federal Specification A-A-1192A (Type 42), ANSI/MSS SP-69 and MSS SP-58 (Type 42).

Ordering: Specify pipe size, material, figure number, name and finish.

Note: If different loads or dimensions are required, refer to Fig. 40 SD special design riser clamp.

Fig. 40: Dimensions (in) Loads (lbs) • Weight (lbs)

<table>
<tr><th rowspan="2">Pipe Size</th><th colspan="2">Max Load</th><th rowspan="2">C-C</th><th rowspan="2">E</th><th rowspan="2">F (max)</th><th rowspan="2">G (max)</th><th rowspan="2">S</th><th rowspan="2">A (CS)</th><th rowspan="2">A (alloy) (SS)</th><th rowspan="2">B (max)</th><th colspan="3">Maximum Weight Each</th></tr>
<tr><th>Rigid Assembly</th><th>Spring Assembly</th><th>CS</th><th>SS</th><th>Alloy</th></tr>
<tr><td>2</td><td rowspan="2">900</td><td rowspan="2">1,800</td><td>18</td><td>9</td><td rowspan="2">⅜</td><td rowspan="2">2½</td><td rowspan="4">¾</td><td rowspan="2">1¼</td><td rowspan="3">⅞</td><td rowspan="8">2</td><td>18</td><td>15</td><td>18</td></tr>
<tr><td>2½</td><td rowspan="2">20</td><td rowspan="2">10</td><td>20</td><td>20</td><td>20</td></tr>
<tr><td>3</td><td>1,500</td><td>3,000</td><td rowspan="3">½</td><td rowspan="2">3</td><td rowspan="2">1½</td><td>30</td><td>25</td><td>30</td></tr>
<tr><td>4</td><td rowspan="2">2,200</td><td rowspan="2">4,400</td><td rowspan="2">22</td><td rowspan="2">11</td><td rowspan="2">1⅛</td><td>40</td><td>40</td><td>44</td></tr>
<tr><td>5</td><td rowspan="2">4</td><td>2</td><td>45</td><td>40</td><td>45</td></tr>
<tr><td>6</td><td rowspan="2">3,000</td><td rowspan="2">6,000</td><td>24</td><td>12</td><td rowspan="2">⅝</td><td rowspan="2">1</td><td rowspan="2">⅞</td><td rowspan="2">1¼</td><td>60</td><td>60</td><td>73</td></tr>
<tr><td>8</td><td>27</td><td>13½</td><td>5</td><td>82</td><td>82</td><td>82</td></tr>
<tr><td>10</td><td>5,500</td><td>11,000</td><td>30</td><td>15</td><td>⅞</td><td>6</td><td>1½</td><td>1¼</td><td>1⅝</td><td>157</td><td>157</td><td>157</td></tr>
<tr><td>12</td><td rowspan="2">7,800</td><td rowspan="2">15,600</td><td>32</td><td>16</td><td rowspan="2">1</td><td rowspan="2">7</td><td rowspan="2">1¾</td><td rowspan="2">1⅜</td><td>1⅞</td><td rowspan="4">2½</td><td>216</td><td>202</td><td>250</td></tr>
<tr><td>14</td><td>34</td><td>17</td><td>1⅜</td><td>228</td><td>228</td><td>263</td></tr>
<tr><td>16</td><td rowspan="2">9,000</td><td rowspan="2">18,000</td><td>36</td><td>18</td><td rowspan="2">1⅛</td><td>6</td><td rowspan="2">2</td><td rowspan="2">1½</td><td rowspan="2">2</td><td>314</td><td>277</td><td>315</td></tr>
<tr><td>18</td><td>39</td><td>19½</td><td>7</td><td>338</td><td>338</td><td>377</td></tr>
<tr><td>20</td><td rowspan="2">13,500</td><td rowspan="2">27,000</td><td>42</td><td>21</td><td rowspan="2">2</td><td>10</td><td rowspan="2">2½</td><td rowspan="2">2⅝</td><td rowspan="2">2⅝</td><td rowspan="2">4¼</td><td>525</td><td>525</td><td>580</td></tr>
<tr><td>24</td><td>45</td><td>22½</td><td>11</td><td>621</td><td>565</td><td>681</td></tr>
</table>

PROJECT INFORMATION	APPROVAL STAMP
Project:	☐ Approved
Address:	☐ Approved as noted
Contractor:	☐ Not approved
Engineer:	Remarks:
Submittal Date:	
Notes 1:	
Notes 2:	